

ABSENCE OF FERROMAGNETIC LONG RANGE ORDER IN RANDOM ISOTROPIC DIPOLAR MAGNETS AND IN SIMILAR SYSTEMS*

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It is shown that random off-diagonal exchange interaction coefficients destroy ferromagnetic long range order in isotropic systems with less than four spatial dimensions. A special case is that of random isotropic systems with dipole–dipole interactions. The proof breaks down when (cubic or hexagonal) symmetry breaking terms are introduced. However, in some cases such terms turn the transition into the ferromagnetic phase first order, with a possible spin glass phase above it.

THE EFFECTS of random impurities on the critical properties of magnets have recently drawn much attention [1]. In some cases, the existence of impurities leads to the drastic result that ferromagnetic long range order, which exists in the absence of impurities, must completely disappear once such impurities are introduced. This has been shown to be the case when random magnetic fields are introduced in systems with rotational invariance (described by Heisenberg or XY models) in $d < 4$ dimensions [2]. Very recently, the same result was shown also to be true [3–5] for systems with random uniaxial anisotropy, i.e. with a single ion term $\sum_i (\mathbf{n}_i \cdot \mathbf{S}_i)^2$ in the Hamiltonian, where $\mathbf{n}_i$ is a unit vector of random direction and $\mathbf{S}_i$ is an n -component spin vector at the site i in the system [6, 7]. The experimental situations with such systems is far from clear, since (a) no magnetic systems with random fields are known and no detailed experiments on other such systems are available, (b) the experiments on amorphous rare earth–iron alloys [6], to which the model of random uniaxial anisotropy is *assumed* to apply, are not very clear, and (c) the assumption of *complete rotational invariance*, needed in the proofs of [2–4], is usually not satisfied. The present paper contains two new results, which are aimed at resolving these difficulties.

First, it is shown that the random uniaxial anisotropy is only a very special case of a large variety of systems in which long range order is eliminated. Consider a general spin system, with the Hamiltonian

$$\mathcal{H} = -\frac{1}{2} \sum_{i,j} \sum_{\alpha,\beta=1}^n J_{ij}^{\alpha\beta} S_i^\alpha S_j^\beta, \quad (1)$$

with random exchange coefficients $J_{ij}^{\alpha\beta}$, such that the configurational averages $[J_{ij}^{\alpha\beta}]_{av}$ maintain the n -dimensional spin space isotropy. We show, that *any random off-diagonal exchange coefficient, $J_{ij}^{\alpha\beta}$ with $\alpha \neq \beta$, will prevent the system from having ferromagnetic long range order for $d < 4$.*

A very important example, which is described by (1) is that of a *random isotropic dipolar magnet*, with [8]

$$J_{ij}^{\alpha\beta} = -(d-2)g_i g_j \mu_B^2 (\delta_{\alpha\beta} - d x_{ij}^\alpha x_{ij}^\beta / r_{ij}^2) / r_{ij}^d, \quad (2)$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $g_i \mu_B$ is the magnetic moment of the ion at site i . Both the moments $g_i \mu_B$ and the locations $\mathbf{r}_i$ may be random, leading to random values of $J_{ij}^{\alpha\beta}$. For isotropic distributions of these random variables, the averages $[J_{ij}^{\alpha\beta}]_{av}$ will describe the usual isotropic dipolar system [8]. The effects of randomness on this system were recently studied, in $d = 4 - \epsilon$ dimensions [9]. It was found, that the isotropic pure dipolar fixed point is *unstable* with respect to randomness. A “random” stable dipolar fixed point was found, but it was not clear whether this fixed point can be reached from physical initial Hamiltonians. Our present results show, that this fixed point *cannot* be reached, since *no ferromagnetic long range order is possible*.

Many other examples exist. For example, quenched random anisotropic strains, $\epsilon_i^{\gamma\delta}$, may induce terms in the Hamiltonian of the form [10]

$$\sum_{\alpha\beta\gamma\delta} g_{\alpha\beta\gamma\delta} \epsilon_i^{\gamma\delta} S_i^\alpha S_i^\beta, \quad (3)$$

leading to similar results. It is hoped that this large variety of possible systems will make a detailed experimental study of the general statements easier.

The second result relates to the assumption of *rotational invariance*. Real crystals usually have symmetry breaking terms, e.g. of cubic or “hexagonal” symmetry [11, 12],

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$$\mathcal{H}_v = -v \sum_i \sum_{\alpha} (S_i^{\alpha})^4, \quad \mathcal{H}_w = -w \sum_i \sum_{\alpha} (S_i^{\alpha})^6. \quad (4)$$

We show, that although such terms make the existence of ferromagnetic long range order possible at low temperatures, *they may have important effects as the transition temperature of the "average" system T_c^0 is approached.* In fact, for some systems *there will be a finite temperature range below T_c^0 in which the long range order is still absent, and the transition into the ferromagnetic ordered phase will have to be first order.* Detailed experimental studies in this range, which can be done in many *real* systems, will contribute much to our understanding of these problems.

There are several ways to demonstrate our statements. First, assume that there is ferromagnetic long range order, with the magnetization M along the 1-axis. Shifting S_i^1 by M , a new term in the Hamiltonian results,

$$\Delta \mathcal{H} = \sum_j \mathbf{H}_j \cdot \mathbf{S}_j, \quad (5)$$

with

$$H_j^{\beta} = M \sum_i (J_{ij}^{1\beta} - [J_{ij}^{1\beta}]_{av}).$$

In fact, we can always choose $\sum_i [J_{ij}^{1\beta}]_{av} = 0$ for $\beta \neq 1$. The term $M \sum_j [J_{ij}^{11}]_{av}$ represents the average demagnetization field, which must indeed be subtracted to yield the correct internal field [13]. The field $\mathbf{H}_j$ is thus a *local random field*, with average zero and with a distribution related to that of $J_{ij}^{1\beta}$. As shown by Imry and Ma [2], there can be no long range ferromagnetic order for $d < 4$. The system will prefer to break into domains, with the local magnetizations parallel to the local random fields. Thus, we must have $M = 0$.

Secondly, consider the fluctuations in one of the transverse magnetization components, $\langle S_j^{\perp} \rangle$. Writing $\langle S_j^{\perp} \rangle = \sum_i G_j^{\perp} H_i^{\perp}$, where $G^{\perp}$ is the transverse susceptibility of the "average" system, and assuming $[(H_j^{\perp})^2]_{av} = \lambda$ (we ignore correlations among the random fields, as these only yield higher order effects), one finds [12]

$$[(S_j^{\perp})^2]_{av} = \lambda \sum_i |G_{ij}^{\perp}|^2 = \lambda (2\pi)^{-d} \int d^d q [G_{\perp}(\mathbf{q})]^2, \quad (6)$$

where $G_{\perp}(\mathbf{q})$ is the Fourier transform of $G_{ij}^{\perp}$. Since the "average" system is isotropic, $G_{\perp}$ diverges as $1/q^2$, and $[(S_j^{\perp})^2]_{av}$ diverges for $d < 4$, which contradicts the finite length of the spin. Transverse spin fluctuations thus eliminate long range order, and $M = 0$. Note that indeed, $G_{\perp}(\mathbf{q})$ behaves as $1/q^2$ for the "average" dipolar case [13]. This can be shown to remain true when the cumulants of the random variables $(J_{ij}^{\alpha\beta} - [J_{ij}^{\alpha\beta}]_{av})$ are included in a perturbation series. Note that this proof works only if $[(H_j^{\perp})^2]_{av} \neq 0$, i.e. if J_{ij}^{12} is random. *The result will not follow if only the diagonal exchange*

coefficients $J_{ij}^{\alpha\alpha}$ are random! In this case, one simply expects a competition between ordering along different axes, and a tetracritical point may arise [14].

We next consider the nature of the phase transition, if any. A ladder approximation, such as used in [3] and [4], yields a *spin glass* ordering. An analysis in $d = 6 - \epsilon$ dimensions, similar to that of Chen and Lubensky [15], then yields exponents which correspond to *Ising-like spin glasses*. Recently, doubts were raised whether this kind of a spin glass exists at all for $d < 4$ [16]. Maybe the spin glass state is observed only for finite measurement times [17]. In such a case, the question of ordering in the systems discussed here remains quite open. It is worth mentioning here that spin glass ordering was also recently predicted for dielectric random dipolar systems, using the molecular field approximation [18], with some support from experiments [19].

We now turn to the effects of the anisotropies, equation (4). As discussed at length in [11] and [12], such anisotropies give rise to a finite transverse susceptibility $1/r_T$, with [11, 20, 21] $r_T \sim v M^{\sigma_v}$. By scaling, $\sigma_v = (\gamma - \phi_v)/\beta$, where ϕ_v is the crossover exponent associated with v . Similar results, with w replacing v , apply when $v = 0$. Replacing $G_{\perp}$ in equation (6) by $1/(q^2 + r_T)$ we now find [12]

$$[(S_j^{\perp})^2]_{av} \propto \lambda r_T^{-\epsilon/2}, \quad (7)$$

where $\epsilon = 4 - d$. In the random field case, discussed by Imry and Ma [2], λ is a number, and (7) becomes $[(S_j^{\perp})^2]_{av} \propto \lambda r_T^{-\epsilon/2} \propto \lambda v^{-\epsilon/2} M^{-\epsilon\sigma_v/2}$. The expression on the right diverges as $M \rightarrow 0$, i.e. as $T \rightarrow T_c^0$. Therefore, the same contradiction discussed earlier arises, and M cannot be non-zero for some temperature range below T_c^0 . Note that if v is very small, this range may be quite wide. Since M may be finite (and of order unity) far below T_c^0 , one must expect a *first order transition* into the ferromagnetic phase. This transition will occur when $\lambda v^{-\epsilon/2} M^{-\epsilon\sigma_v/2}$ becomes of order unity. A spin glass phase may still be observed for some temperature range above this transition.

In the random off-diagonal exchange (or anisotropy) case, $H_i^{\perp} \propto M$ [equation (5)], and hence $[(S_j^{\perp})^2]_{av} \propto \Delta v^{-\epsilon/2} M^{2-\epsilon\sigma_v/2}$, where Δ is proportional to the square of the width of the random distribution of J_{ij}^{12} . The first order transition is therefore still expected when $2 - \epsilon\sigma_v/2 < 0$, or $\phi_v < \gamma - 4\beta/\epsilon$, and it will take place when $\Delta v^{-\epsilon/2} M^{2-\epsilon\sigma_v/2} \sim 1$. For isotropic short range Heisenberg models, the exponents β and γ have very good estimates at $d = 3$ [22]. For both $n = 2$ and $n = 3$ one has $\gamma - 4\beta/\epsilon \approx -0.07$. The exponent ϕ_v is very difficult to estimate at $d = 3$ [21], and it is very close to zero for $n = 3$, and more negative for $n = 2$. The condition for a first order transition is probably satisfied for $n = 2$, and not for $n = 3$. In any case,

$(2 - \epsilon\alpha_v/2)$ is very small, so the expected effect is small. The situation for dipolar interactions is even less clear, as only short ϵ -expansions are available [23]. At $n = d = 4 - \epsilon$, these yield $\phi_v > \gamma - 4\beta/\epsilon$. We thus see that in most cases, a sufficiently large cubic perturbation $\mathcal{H}_v$ will eliminate the effect. However, for XY-like systems or for small v the effect is still expected [24].

One should note that the parameter v is characteristic of the *average* Hamiltonian. By changing the details of the randomness one can thus try to make v very small. If v is sufficiently small, the next order anisotropy $\mathcal{H}_w$ becomes important, and the same results are found with w replacing v everywhere. The exponent ϕ_w is negative and large (for $n = 2$, $\phi_w \simeq -1.58$ [12]), so that *always* $\phi_w < \gamma - 4\beta/\epsilon$, and the magnetization cannot become too small below the transition.

As mentioned in the beginning, the experimental situation is far from clear. Amorphous rare-earth alloys, or similar materials to which the random uniaxial anisotropy applies, do seem to show a region of spin glass behavior [4], and probably become ferromagnetic at

lower temperatures. The random cubic ferroelectric $\text{KTa}_{0.9}\text{Nb}_{0.1}\text{O}_3$ does exhibit a ferroelectric ordering, with unusual critical exponents [25]. Could these unusual effects be related to some narrow intermediate "spin glass" phase? The standard examples of isotropic dipolar ferromagnets are EuS and EuO [26]. Experiments on EuO_{1-x} have shown [27] a decrease in the transition temperature for $x \neq 0$, and some "smearing" which may again be related to an intermediate phase. More recently, detailed investigations of the phase diagram of $\text{Eu}_{1-x}\text{Sr}_x\text{S}$ have discovered regions of spin glass ordering [28], which may be interpreted as realizations of the effects predicted in this paper. It is hoped that this paper will stimulate additional detailed experiments on dipolar amorphous systems.

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